Rocuronium

****This monograph is under review****

Neonatal

FORM Vial containing 50mg in 5ml

INDICATION Muscle Relaxation **(ONLY USE WHEN VECURONIUM NOT AVAILABLE)**

DOSE RANGE

SLOW IV INJECTION

AGE	DOSE	FREQUENCY	ROUTE
Neonate	600micrograms/kg	Initial dose then further doses or continuous infusion (see below)	IV
Neonate	100-150micrograms/kg	Every 2 to 4 hours as necessary	IV

CONTINUOUS INFUSION

AGE	DOSE	FREQUENCY	ROUTE
0 – 6 months	Starting dose 300micrograms/kg/hour Dose can be increased by 100micrograms/kg/hour increments if necessary for adequate clinical effect to a maximum of 600micrograms/kg/hour	Continuous Infusion (see below)	IV

NB. Daily Rocuronium 'holidays' should be considered.
To achieve a holiday, stop the infusion until evidence of reversal of neuromuscular blockade is observed (e.g. spontaneous breathing or any other movement), then restart without increasing original dose

Aim to use the lowest possible dose & regular Rocuronium holidays to obtain a desired level of blockade while limiting risk of adverse effects or prolonged blockade.

PRESCRIPTION OF CONTINUOUS INFUSION

50mg/kg in 50ml infusion fluid
This gives:-

- 300microgram/kg/hour at 0.3ml/hr
- 600microgram/kg/hour 0.6ml/hr

RECONSTITUTION Already in solution

DILUTION

For Slow IV injection

Rocuronium 10mg/ml inj.	1ml
Glucose 5%	Up to 10ml total

Gives a 1mg/ml solution. Use the required volume.

For continuous Infusion

Using 50mg in 5ml **injection**
5 x wt (kg) is the number of ml of 50mg in 5ml to be diluted up to 50ml total with infusion fluid (equivalent to **50mg/kg in 50ml**)

METHOD OF ADMINISTRATION

Slow IV injection
Bolus doses can be given as a separate bolus (pg 1) or given from the continuous infusion syringe (above) e.g. 0.6ml (600micrograms/kg) over 5 minutes.
(PRM use 600 microgram/kg and Hands Free bolus to give this IV bolus at 10 ml/hr over 3 minutes 36 seconds)

For continuous infusion
By continuous intravenous infusion, flow rate adjusted according to the baby's response (see prescription section for details).

COMPATIBILITY

Solution compatibility	Glucose 5%, Sodium Chloride 0.9%
Solution incompatibility	No info
IV Line compatibility	Amiodarone, Aprotinin, Milrinone, Morphine,
IV Line incompatibility	Furosemide, Insulin, Vancomycin

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS
See Summary of Product Characteristics and most recent edition of BNF for Children
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FURTHER INFORMATION

- Onset of action is usually within 2minutes with intermediate duration of action.
- Onset of action is shortened and duration is prolonged as the dose increases.
- Prolonged neuromuscular blockade may be reversed by use of sugammadex (seek advice).
- Neonates have an increased sensitivity to non-depolarising muscle relaxants-care is required

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LICENSED STATUS Unlicensed

APPLICABLE POLICIES West of Scotland Neonatal Guidelines:

Consult local policy if applicable

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Network. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department